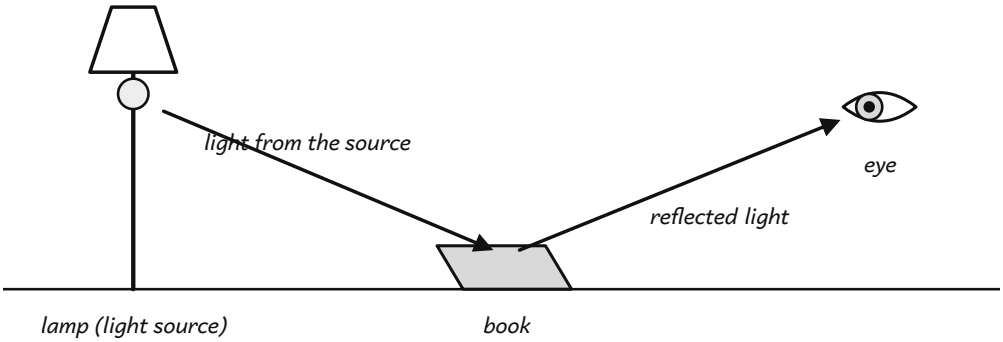
PHYSICAL AND CHEMICAL SCIENCE

Light: Sources, Reflection and Refraction

How we see, mirrors and reflection, refraction, optical devices and the human eye



In this pack

- **Learn it:** Light: how we see things ★☆☆
- **Fact check:** Light sources, mirrors and the eye ★☆☆
- **Think it through:** Tracing the path of light ★★☆
- **Investigate:** Investigating mirrors: angles of reflection ★★☆
- **Real-world case study:** Healthy eyes in a screen world ★★★
- **Review and challenge:** Show what you know ★★★

This pack belongs to: _____ **Class:** _____

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LEARN IT · C7 · LIGHT AND SEEING

Light: how we see things

★☆☆
Grade 5

Name: Class: Date:

Learning goals

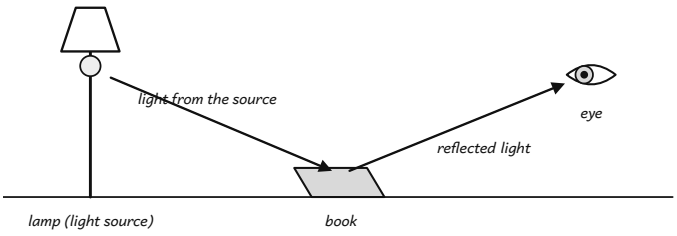
- Explain how we see light sources and other objects.
- Describe reflection, lateral inversion and refraction.
- Describe how optical devices and the human eye work.

Close your eyes and you see nothing. Open them and the world appears, full of colour and detail. What is happening? Scientists have argued about this for more than two thousand years. Today we know the answer: **we see when light enters our eyes**.

Light sources

A **light source** makes its own light: the Sun, a candle flame, a light bulb, a torch, lightning and even a firefly (*jonaki*). Most things are **not** light sources. A book, a tree, your friend's face and even the **Moon** only **reflect** light that falls on them. The Moon looks bright at night because it reflects light from the Sun.

We see a light source when its light travels **straight into our eyes**. We see every other object when light from a source **bounces off** (reflects from) the object and then enters our eyes.



How we see a book.

⚠ Stay safe

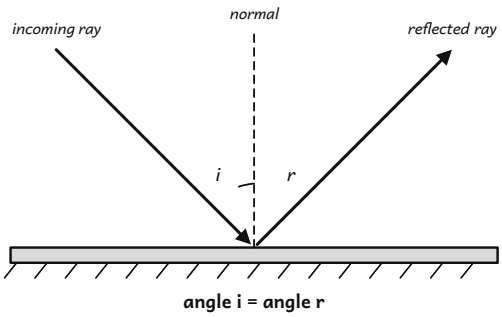
Never look directly at the Sun, even with sunglasses. Its light is strong enough to damage your eyes for ever.

Absorbed, reflected or refracted

When light hits a material, three things can happen. Dark, dull materials **absorb** most of the light, so they look dark. Light-coloured materials **reflect** a lot of light. Transparent materials such as glass and water let light pass through, but they can bend it (**refract** it).

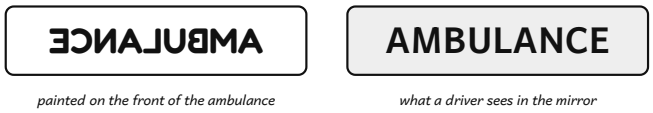
Reflection and mirrors

A **mirror** is very smooth and shiny, so it reflects light neatly in one direction. A ray of light hits the mirror and bounces off at the **same angle**: the angle of the incoming ray equals the angle of the reflected ray, measured from a line at right angles to the mirror called the **normal**. Rough surfaces scatter light in all directions, so they do not make a clear image.



The law of reflection.

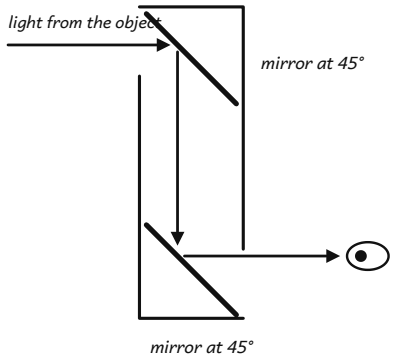
Your image in a mirror is the right way up but **back to front**. Raise your right hand, and the image raises its left. This is called **lateral inversion**. That is why the word AMBULANCE is painted back to front on the front of ambulances: drivers in front read it the right way round in their mirrors.



painted on the front of the ambulance *what a driver sees in the mirror*

Lateral inversion.

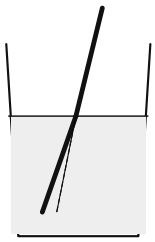
A **periscope** uses two mirrors at 45° to let you see over a wall or round a corner. Submarines use them to see above the water.



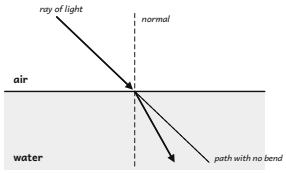
Light reflects twice inside a periscope.

Refraction

Light travels more slowly in water and glass than in air. When light passes from one material into another at an angle, it **changes direction**. This is **refraction**. It makes a pencil in a glass of water look broken, and it makes ponds look shallower than they really are.



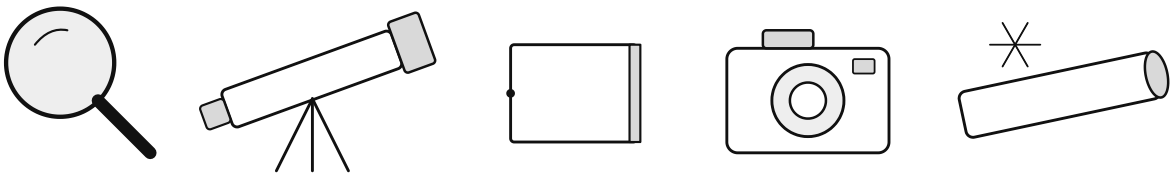
A pencil looks bent.



Light bends as it enters water.

Optical devices

Device	What it does to light
hand lens	a curved glass **lens** refracts light to make things look **bigger** (magnified)
telescope	lenses or curved mirrors collect light from far away so stars and planets look **closer**
pinhole camera	light travels in straight lines through a tiny hole to make an **upside-down** image
camera	a lens **focuses** light onto a sensor (or film) to record an image
kaleidoscope	mirrors inside a tube reflect coloured pieces again and again to make **repeating patterns**

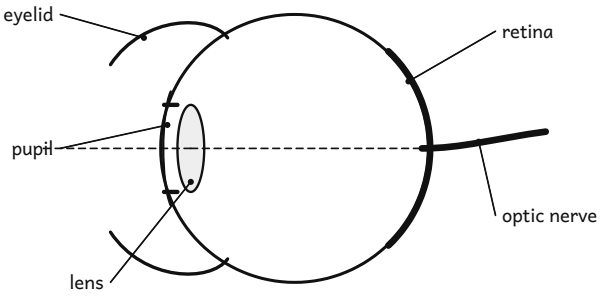


hand lens **telescope** **pinhole camera** **camera** **kaleidoscope**

Five optical devices.

The human eye

Your **eyelid** protects the eye and spreads tears to keep it clean. Light enters through the **pupil**, a hole that gets smaller in bright light and bigger in the dark. The **lens** focuses the light to make a tiny upside-down image on the **retina** at the back of the eye. Millions of light-sensitive cells in the retina send messages along the **optic nerve** to the **brain**, which turns the image the right way up and tells you what you are seeing.



Inside the eye (side view).

From the past

Around the year 1015, the scientist **Ibn al-Haytham** (born in Basra, worked in Cairo) wrote the *Book of Optics*. Through careful experiments, including with a darkened room and a tiny hole (a pinhole camera), he showed that we see because light **enters** our eyes, not because our eyes send out rays, as many people believed. He is often called the father of optics.

Key words

light source	something that makes its own light
reflect	bounce light off a surface
absorb	take in light
refraction	light changing direction as it passes into a different material
normal	a line at right angles to a mirror or surface
lateral inversion	a mirror image is back to front
lens	curved glass or plastic that bends light
retina	the light-sensitive layer at the back of the eye

FACT CHECK · C7 · LIGHT AND SEEING

Light sources, mirrors and the eye

★☆☆
Grade 5

Name: Class: Date:

Learning goals

- Identify light sources.
- Label the eye.
- Recall what optical devices do.

1 Sort these objects. [8]

SORT THESE

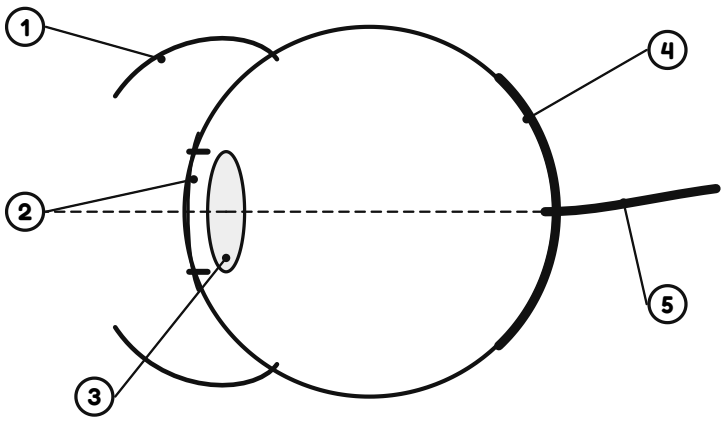
a white page **an LED bulb** **a candle flame** **the Sun** **a shiny spoon** **a firefly** **a mirror** **the Moon**

Light source	Not a light source (reflects light)

2 Label the parts of the eye. [5]

LABELS

eyelid **pupil** **retina** **lens** **optic nerve**



1 _____ 2 _____ 3 _____

4 _____ 5 _____

3 Match each optical device to what it does. [5]

Draw a line to match each pair, or write the letter.

hand lens ●	● **A** makes an upside-down image through a tiny hole
telescope ●	● **B** uses mirrors to make repeating patterns
pinhole camera ●	● **C** uses two mirrors to see over a wall
kaleidoscope ●	● **D** makes far-away objects look closer
periscope ●	● **E** makes small things look bigger

4 True or false? [4]

Statement	True	False	If false, correct it
1. The Moon is a light source.			
2. Light hits a mirror and bounces off at the same angle.			
3. Our eyes send out rays so that we can see.			
4. The pupil gets bigger in the dark.			

Tracing the path of light

★★☆
Grade 5

Name: Class: Date:

Learning goals

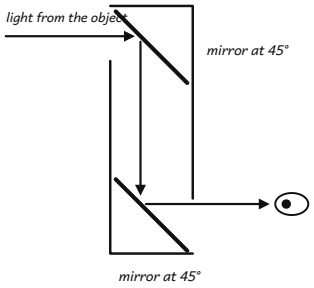
- Draw ray diagrams.
- Explain reflection, refraction and lateral inversion.

1 **Draw arrows to show how Mina sees the mango. Label the light source.** [2]

Sun → mango → Mina's eye

2 **A ray of light hits a mirror at an angle of 35° to the normal. What is the angle of the reflected ray? Explain.** [2]

3 **Look at the periscope.** [3]



a) Why must both mirrors be at 45°?

b) Give one real use of a periscope.

4 **Rafi writes his name on a card and holds it up to a mirror. Describe what he sees and use the scientific term.** [2]

5 **A fisherman spears a fish that looks close to the water surface, but he misses. Use refraction to explain why.** [2]

6 **Why does your pupil get smaller when you walk out into bright sunshine?** [1]

Investigating mirrors: angles of reflection

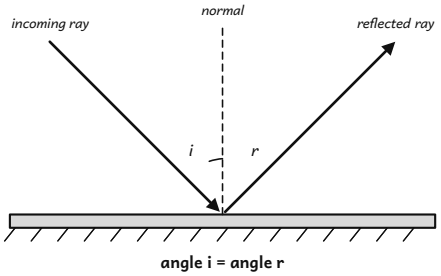
★★☆ Grade 5

Name: Class: Date:

Learning goals

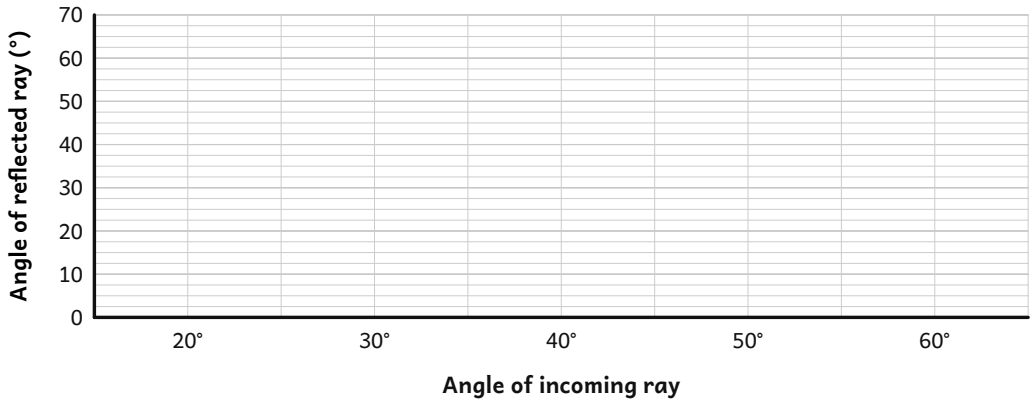
- Measure angles with a protractor.
- Spot a result that does not fit.
- Draw a conclusion.

Class 5C shone a thin beam of light from a ray box onto a plane mirror. They drew the normal, then used a protractor to measure the angle of the incoming ray and the angle of the reflected ray.



1 **Here are the results.** [5]

Angle of incoming ray (°)	20	30	40	50	60
Angle of reflected ray (°)	20	31	40	56	60



a) Plot the results as a line graph.

b) Which result does **not** fit the pattern?

.....

c) Suggest a reason for this result and what the class should do.

.....

.....

2 **Write *a* conclusion.** [2]

3 **Predict the *a*ngle of reflection for *a*n incoming r*a*y *a*t 75°.** [1]

4 **Why did they use *a* thin beam of light rather than *a* wide torch beam?** [1]

REAL-WORLD CASE STUDY · C7 · LIGHT AND SEEING

Healthy eyes in a screen world

★★★
Grade 5

Name: **Class:** **Date:**

Learning goals

- Use data from a survey.
- Link eye health to the structure of the eye.
- Plan a health campaign.

Science · Health · Maths · Technology · Wellbeing

More and more children around the world are becoming **short-sighted** (myopic): they see near things clearly, but far things look blurred, because the image forms in front of the retina instead of on it. Scientists estimate that by 2050 about **half of the world's people** could be short-sighted.

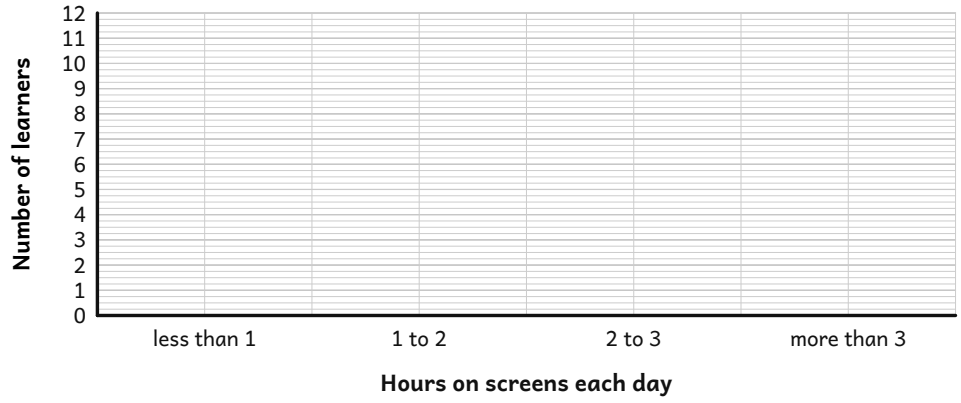
Research suggests that spending time **outdoors in daylight** helps to protect children's eyes. In Taiwan, schools encouraged children to spend 120 minutes outdoors every day, and the number of short-sighted children began to fall. Eye doctors also recommend the **20-20-20 rule**: every 20 minutes of close work or screen use, look at something about 6 metres (20 feet) away for 20 seconds.

Food matters too. Without enough **vitamin A**, people can develop **night blindness**. Bangladesh runs a national Vitamin A campaign that gives capsules to young children, and vitamin A comes from foods such as spinach, carrots, mangoes, eggs and small fish like *mola*.

Based on research reported in the journal Ophthalmology (2016) and Taiwan school studies; Bangladesh National Vitamin A Plus Campaign.

1 **Class 5C did a survey of 30 learners: "How many hours do you spend on screens each day, outside school?" (example data)** [3]

Hours per day	less than 1	1 to 2	2 to 3	more than 3
Number of learners	4	9	10	7



a) Draw the bar chart.

b) How many learners spend 2 hours or more on screens each day?

.....

2 **Use your knowledge of the eye to explain why a short-sighted person sees far objects as blurred.** [2]

3 **What might each person say about screens?** [3]

Person	View
a learner	
a parent	
an eye doctor	

4 **Design a "Healthy Eyes Week" for your school. Give three actions and the reason for each.** [3]

Show what you know

★★★ Grade 5

Name: Class: Date:

Learning goals

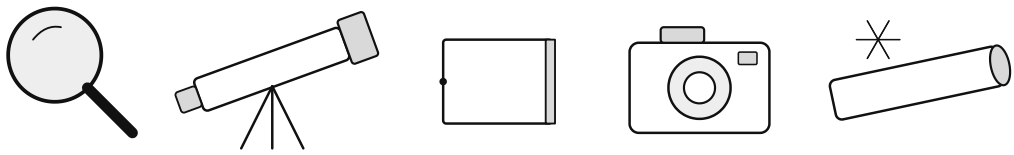
- Use key vocabulary.
- Answer test-style questions.

1 Find the key words.

L	C	R	E	T	I	N	A	L	P	U
F	O	D	E	R	N	S	J	A	Y	R
U	U	M	I	R	R	O	R	A	R	T
F	H	U	G	O	E	U	J	E	F	M
D	C	N	C	A	F	R	T	J	J	O
Y	W	H	M	M	L	C	Y	E	C	V
D	V	Y	T	K	E	E	A	P	L	N
B	H	G	L	W	C	C	K	U	I	E
B	L	E	N	S	T	A	B	P	G	R
I	M	A	G	E	F	M	M	I	H	V
Y	O	A	H	P	M	M	E	L	T	E

image lens light mirror nerve pupil reflect retina source

2 Write the letter of each optical device. [5]



A B C D E

Device	Letter
camera	
kaleidoscope	
hand lens	
pinhole camera	
telescope	

3 Which part of the eye has light-sensitive cells? [1]

- ◯ eyelid
- ◯ pupil
- ◯ retina
- ◯ lens

4 **A straw in a glass of water looks bent. This is because of …** [1]

- ◯ reflection
- ◯ refraction
- ◯ absorption
- ◯ lateral inversion

5 **Challenge. Explain how you can see this page. Use the words *source*, *reflect*, *pupil*, *retina*, *nerve* and *brain*.** [3]

How am I doing? Colour or tick a circle.

I can…	Not yet	Nearly	Got it!
explain how we see light sources and other objects	◯	◯	◯
use the law of reflection	◯	◯	◯
explain lateral inversion and how a periscope works	◯	◯	◯
explain refraction with an example	◯	◯	◯
describe what five optical devices do	◯	◯	◯
explain how the eye works and how to keep eyes healthy	◯	◯	◯